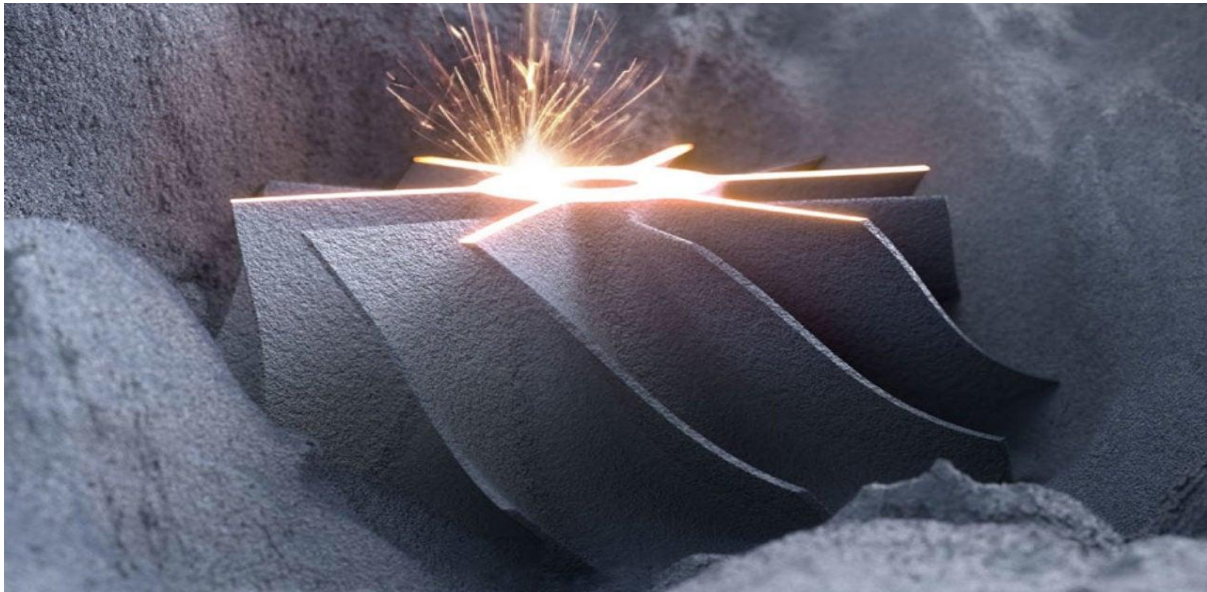




**COURSE CONTENT**

- 2.1.1 Introduction**
- 2.1.2 Selective Laser Sintering (SLS)**
- 2.1.3 SLS Materials**
- 2.1.4 Comparison of Solid, Liquid and Powder based AM**
- 2.1.5 3D Printing / Binder Jetting**
- 2.1.6 3DP Materials**
- 2.1.7 Laser Engineered Net Shaping (LENS)**
- 2.1.8 LENS Materials**
- 2.1.9 Applications of LENS**

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### 2.1.1 INTRODUCTION

Powder-based AM processes represent one of the most widely adopted approaches for producing complex and high-performance components directly from digital models. These processes rely on fine powders of metals, polymers, or ceramics as the raw material, which are selectively fused or deposited to build parts layer by layer. Owing to their capability to produce intricate geometries, high material utilization, and suitability for both prototyping and functional end-use applications, powder-based AM processes have gained significant industrial importance.

Broadly, powder-based AM processes are classified into two main categories:

1. Powder Bed Fusion (PBF): In PBF processes, a thin layer of powder is spread across the build platform, and thermal energy sources such as a laser or an electron beam are employed to selectively fuse the powder according to the cross-sectional geometry of the part. Initially developed for fabricating plastic prototypes using laser scanning, PBF has since evolved to include metals and ceramics, offering high precision, fine resolution, and excellent surface finish.

Various AM systems fall under this category are:

- Selective Laser Sintering (SLS) – Fuses polymer powders (e.g.: Nylon).
- Selective Laser Melting (SLM) – Fully melts metal powders (e.g., titanium, stainless steel).
- Direct Metal Laser Sintering (DMLS) – Similar to SLM, often used for alloys.
- Electron Beam Melting (EBM) – Uses electron beams to melt metal powders (commonly titanium).
- Multi Jet Fusion (MJF) – Uses fusing and detailing agents and infrared heating on polymer powders.

2. Binder Jetting: In Binder Jetting processes, a thin layer of powder is spread across the build platform, and a liquid binder is selectively deposited using an inkjet printhead to join powder particles according to the cross-sectional geometry of the part. The process is carried out at room temperature, unlike PBF, and does not require thermal energy to fuse the material. Binder Jetting is particularly suitable for rapid concept models, full-color prototypes, and low-cost production of metal or ceramic parts after post-processing such as sintering or infiltration.

Various AM systems fall under this category:

- Z Corporation's 3DP – Uses plaster or gypsum-based powders with colored binder for full-color concept models.
- Voxeljet – Large-scale binder jetting systems for sand molds and architectural models.

3. Directed Energy Deposition (DED): In contrast, DED processes involve the simultaneous feeding of powder (or) sometimes wire material and the application of a focused thermal energy source - commonly a laser, electron beam, (or) plasma arc to melt the material as it is deposited. The process is highly flexible and is particularly suitable for repairing, adding features to existing components, and producing large, near-net-shape parts.

AM systems under DED are:

- Laser Engineered Net Shaping (LENS) – Delivers metal powder into a Laser beam for melting.
- Electron Beam Additive Manufacturing (EBAM) – Uses an Electron beam to melt metal wire feedstock.
- Wire Arc Additive Manufacturing (WAAM) – Uses an Electric arc to melt wire material.
- Direct Metal Deposition (DMD) – Similar to LENS, Laser-based with powder feeding.

In the present chapter, we will focus on one representatives of these categories: SLS under PBF, 3DP under Binder Jetting and LENS under DED. These processes will be discussed in detail with respect to their working principles, materials, advantages, limitations, and applications.

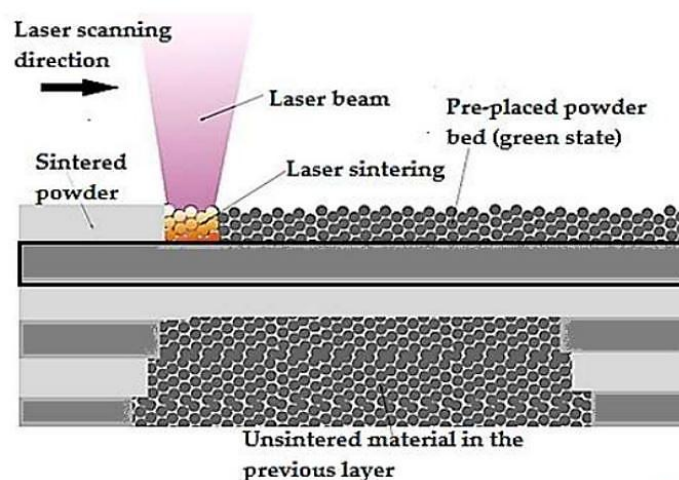
### 2.1.2 SELECTIVE LASER SINTERING (SLS)

SLS was developed at the University of Texas at Austin and commercialized by DTM Corporation in 1987 with support from B.F. Goodrich. DTM held the worldwide exclusive license to market SLS technology until the company was acquired by 3D Systems in August 2001.

In SLS, a part is built layer by layer by selectively sintering powdered material with a high-powered CO<sub>2</sub> laser. A thin layer of powder is spread onto the build platform, and the laser scans the cross-sectional geometry of the CAD model. The laser energy raises the powder temperature above its glass-transition point, softening particles enough for them to bond together and to the previous layer. This process is repeated until the entire part is completed.

#### SINTERING

Sintering involves heating powder particles above their glass-transition temperature, where the material softens but does not fully melt. In this semi-solid state, particles deform at their contact points, allowing them to bond and form a solid structure without entering a fully molten phase. This reduces the risk of distortion or warping that can occur with complete melting. After cooling, the bonded powder forms a stable matrix with a density close to that of the raw material.

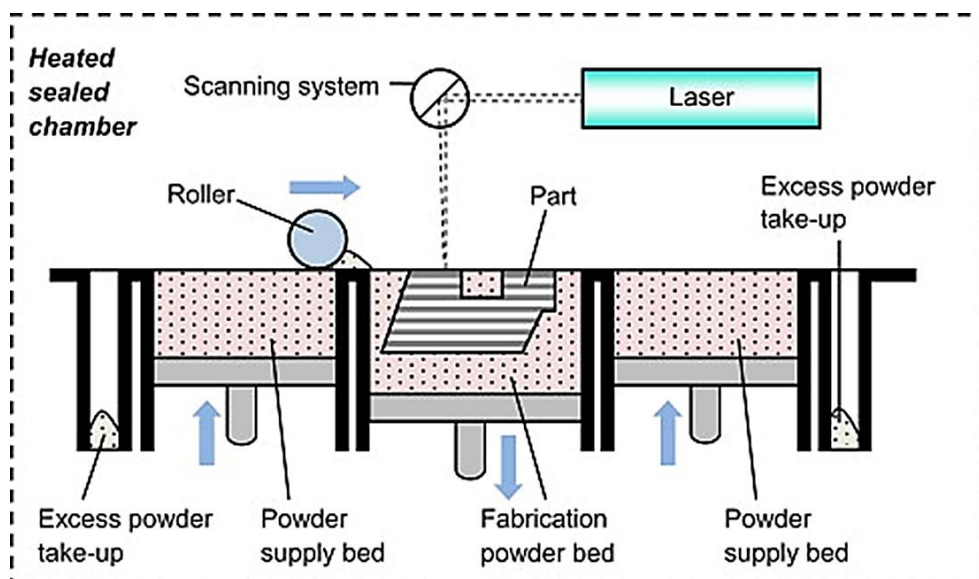


***Principle of Sintering***

Because the process requires heating the particles to near their transition temperature, SLS consumes significantly more energy approximately 300 to 500 times higher than photopolymerization processes. To reduce laser power requirements, auxiliary heaters preheat the powder bed close to the sintering temperature. An inert gas environment (such as nitrogen) is essential to prevent oxidation or explosion of fine powders, and chamber cooling systems are used to stabilize the build environment.

### HARDWARE

- Fabrication Powder Bed: Also referred to as Build Platform where the part is built layer by layer. It moves in – Z direction after printing each layer.
- Powder Supply Beds: Two reservoirs on either side of the build platform that move in the +Z direction to feed fresh powder.
- Excess Powder Take-up Chamber: Collects surplus powder displaced during recoating.
- Roller: Spreads a thin, uniform powder layer from the supply bed onto the build platform.
- CO<sub>2</sub> Laser: Provides the energy required to sinter the powder particles.
- Optical Scanning System: Directs and controls the laser beam path with precision, guided by the CAD model.



***Hardware and Principle of SLS***

### PROCESS

A 3D CAD model in STL file format after slicing is sent to the machine for building the part layer by layer. A thin layer of heat-fusible powder is deposited onto the part building chamber. The bottom-most cross-sectional slice of the CAD part under fabrication is selectively drawn on the layer of powder by a heat-generating CO<sub>2</sub> laser. The interaction of the laser beam with the powder elevates the temperature to the point of melting, fusing the powder particles to form a solid mass. When the cross-section is completely drawn, an additional layer of powder is deposited via a roller mechanism on top of the previously

scanned layer. The process is repeated, with each layer fusing to the layer below it and until the part is completed.

The unfused powder during processing serves as a built-in support structure. There is no need to create support structures and thus no support structure to remove when the part is completed.

After the part is built, the powder cake is taken to the Breakout station table, where the excess powder is removed from the parts manually with air brushes and vacuum. The parts may then require some post-processing (or) secondary finishing, such as sanding, lacquering and painting, depending upon the application of the prototype built.

#### ADVANTAGES:

1. **Good part stability:** Parts are created within a precise controlled environment. The process and materials provide for directly produced functional parts to be built.
2. **Wide range of processing materials:** A wide range of materials including Nylon, TPU, TPE thus providing flexibility and a wide scope of functional applications.
3. **No part supports required:** The system does not require CAD developed support structures.
4. **Little post-processing required:** The finishing of the part is reasonably fine and requires only minimal post-processing such as particle blasting and sanding.
5. **No post-curing required:** The completed laser sintered part is generally solid enough and does not require further curing.
6. **Batch production capability:** Multiple parts can be produced simultaneously.

#### LIMITATIONS:

1. **Large physical size of the unit:** The system requires a relatively large space to house it. Apart from this, additional storage space is required to house the inert gas tanks used for each build.
2. **High power consumption:** The system requires high power consumption due to the high wattage of the laser required to sinter the powder particles together.
3. **Surface finish limitations:** As-built parts often have a rougher texture due to powder granularity.

### 2.1.3 SLS MATERIALS

In SLS, a wide range of materials can be used depending on the intended application, ranging from polymers for functional prototypes to metals and ceramics for high-performance and specialized parts. Polymers, particularly Nylon and its reinforced composites, are the most commonly used due to their balance of strength, toughness, and processability. Metals are processed using SLS variants such as DMLS and SLM to produce dense, functional, and load-bearing components, while ceramics are primarily explored for high-temperature, wear-resistant, and biomedical applications. The following table categorizes these materials along with their key properties and typical applications.

***SLS Materials – Composition, Properties, and Applications***

| Category        | Material           | Key Properties                            | Applications                                      |
|-----------------|--------------------|-------------------------------------------|---------------------------------------------------|
| <b>Polymers</b> | Nylon (PA11, PA12) | High strength, toughness, wear & chemical | Automotive components, housings, tooling, medical |

|                 |                                                                                      | resistance                                                    | models                                                   |
|-----------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------|----------------------------------------------------------|
|                 | Glass Filled Nylon                                                                   | High stiffness, dimensional stability, wear resistance        | Structural parts, enclosures, jigs & fixtures            |
|                 | Aluminum Filled Nylon                                                                | Metallic look, high stiffness, thermal stability              | Automotive prototypes, functional & heat-resistant parts |
|                 | Carbon Fiber Reinforced Nylon                                                        | High stiffness-to-weight ratio, excellent strength            | Aerospace, motorsport, lightweight tooling               |
|                 | PEEK / PEKK                                                                          | High strength, thermal & chemical resistance, biocompatible   | Aerospace, high-performance parts, medical implants      |
|                 | Thermoplastic Polyurethane (TPU)                                                     | Flexibility, elasticity, impact resistance                    | Footwear, seals, gaskets, flexible prototypes            |
| <b>Metals</b>   | Stainless Steel                                                                      | High strength, corrosion resistance                           | Medical tools, aerospace parts, functional prototypes    |
|                 | Aluminum                                                                             | Lightweight, strength-to-weight ratio, thermal conductivity   | Automotive, aerospace, heat exchangers                   |
|                 | Titanium                                                                             | High strength, lightweight, biocompatibility                  | Medical implants, aerospace structures                   |
|                 | Cobalt Chrome                                                                        | Wear resistance, biocompatibility, high-temperature stability | Dental implants, turbine blades, orthopedic parts        |
|                 | Nickel Alloys                                                                        | High-temperature strength, oxidation & corrosion resistance   | Jet engines, gas turbines, heat exchangers               |
| <b>Ceramics</b> | Silica (SiO <sub>2</sub> )                                                           | High-temp resistance, insulating                              | Investment casting molds, heat shields                   |
|                 | Alumina (Al <sub>2</sub> O <sub>3</sub> )                                            | Hardness, wear resistance, electrical insulation              | Cutting tools, medical implants, substrates              |
|                 | Zirconia (ZrO <sub>2</sub> )                                                         | High toughness, thermal shock resistance                      | Dental crowns, wear parts, cutting inserts               |
|                 | Hydroxyapatite (Ca <sub>10</sub> (PO <sub>4</sub> ) <sub>6</sub> (OH) <sub>2</sub> ) | Biocompatible, bone-like                                      | Bone scaffolds, biomedical implants                      |

#### 2.1.4 COMPARISON OF SOLID, LIQUID, AND POWDER BASED AM

AM encompasses a variety of technologies that differ in the state of the material as solid, liquid, or powder. Solid-based processes, such as FDM, extrude thermoplastic filaments to build parts layer by layer. Liquid-based processes, like SLA, cure photopolymer resins using light to create high-resolution models. Powder-based processes, such as SLS, fuse powdered materials using thermal energy to produce functional prototypes and end-use parts. Each method has distinct material requirements, process characteristics, and application areas, making it important to compare them in terms of material type, mechanical properties, and typical uses.

The following table shows the Comparison of Solid, Liquid, and Powder Based AM Technologies: FDM, SLA, and SLS

**Comparison of FDM, SLA, and SLS AM Technologies**

| <b>Criteria</b>                            | <b>FDM</b>                                                            | <b>SLA</b>                                                                                                          | <b>SLS</b>                                                                       |
|--------------------------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>Material</b>                            | Thermoplastic filaments: PLA, ABS, PETG, Nylon, PC etc.               | Photopolymer resins (acrylic or epoxy based)                                                                        | Powdered polymers Nylon, TPU powders and ceramics.                               |
| <b>Layer Thickness (Z-axis resolution)</b> | 0.1 – 0.3 mm (100–300 $\mu\text{m}$ )                                 | 0.025 – 0.1 mm (25–100 $\mu\text{m}$ )                                                                              | 0.06 – 0.15 mm (60–150 $\mu\text{m}$ )                                           |
| <b>Surface Finish</b>                      | Noticeable layer lines; stepped effect especially on curved surfaces  | Very high detail; smooth surfaces; sharp features                                                                   | Surface is grainy/matte because of powder particle sintering;                    |
| <b>Support Structures</b>                  | Required for overhangs, support removal and surface finishing needed. | Required; supports are usually fine, support removal and post curing.                                               | Generally not required: unsintered powder around the part acts as support.       |
| <b>Strength</b>                            | Good, but parts are weaker in layer bonding direction.                | Resins tend to be more brittle; strength lower than compared to FDM (or) SLS                                        | Generally stronger. Better impact resistance, thermal performance, durability.   |
| <b>Dimensional Accuracy / Precision</b>    | Less precision: typically, tolerance is $\pm 0.2 - 0.5$ mm            | Higher precision: SLA can achieve tighter tolerances $\pm 0.05$ mm                                                  | Moderate precision: typical tolerances around $\pm 0.1 - 0.3$ mm,                |
| <b>Build Speed</b>                         | Moderate; Complex geometries or large parts take time                 | SLA is slow; DLP (or) MSLA variants can cure whole layers simultaneously, but cleaning, post curing add extra time. | SLS can be efficient for batch production. Also no supports so less wasted time. |
| <b>Cost (Machine + Materials)</b>          | Lowest among the three                                                | Medium to high                                                                                                      | Highest                                                                          |

**2.1.5 BINDER JETTING / 3D PRINTING (3DP)**

The technology is often referred to as 3DP technology and is copyrighted under this name, in which a binder is printed onto a powder bed to form part cross sections.

Binder Jetting is an additive manufacturing process in which an industrial printhead selectively deposits a liquid binding agent onto a thin layer of powder particles such as foundry sand, polymer and ceramics to build parts and tooling.

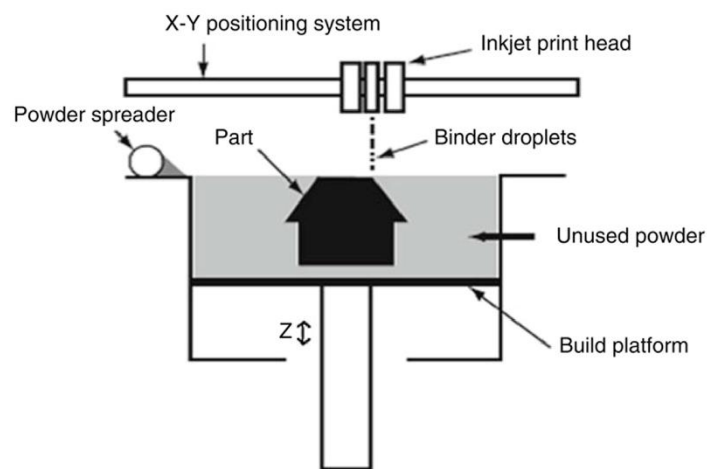
3DP process was developed at the Massachusetts Institute of Technology (MIT) in the early 1990s and was licensed to Z-Corporation for commercialization in 1995. The company became well-known for producing rapid, low-cost concept models, particularly due to its unique capability of printing full-color prototypes. Z Corporation was later acquired by 3D Systems in 2012.



In 3DP, a part is built layer by layer by selectively depositing a liquid binder onto a thin layer of powdered material. An inkjet printhead passes over the powder bed, jetting droplets of binder to glue the particles together in the cross-sectional geometry of the CAD model. Colored inks (CMYK) can be added to the binder to produce vivid, multi-colored models. The process is repeated until the entire part is completed.

### BINDING

Since the process relies on inkjet binder deposition, energy requirements are significantly lower compared to processes such as SLS. No lasers, heaters, or inert gas chambers are required. The powder bed itself provides natural support for the part during printing, eliminating the need for additional support structures. This makes the process faster and more economical, but the produced models are generally weaker and more porous compared to laser-sintered parts.



***Principle of Binding***

### HARDWARE

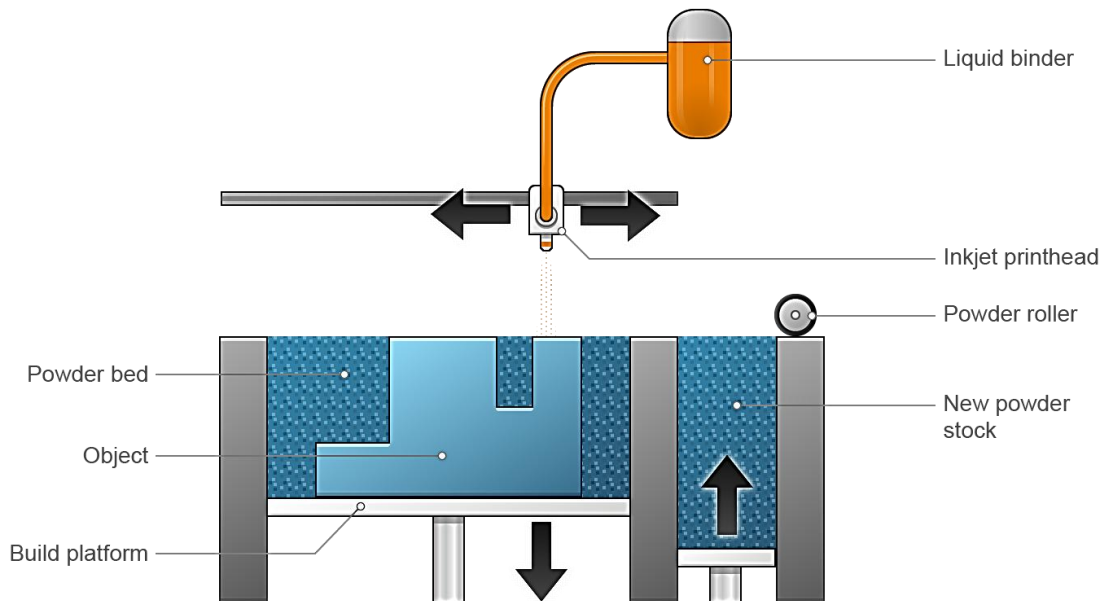
- **Build platform**: Also referred as Powder Bed where the part is constructed layer by layer on this platform, which moves in  $-Z$  direction after each layer is printed.
- **Powder Feed bed**: A supply reservoir that holds the raw powder and raises in  $+Z$  direction to supply new material.
- **Powder roller**: Spreads a thin, uniform layer of powder from the feed bed onto the build platform.
- **Inkjet Printhead**: Jets binder droplets and colored inks onto the powder surface to define the part geometry.

### PROCESS

A CAD model is first converted into an STL file and sliced into thin layers. The printing process begins with a thin layer of powder being uniformly spread across the build platform. An inkjet printhead then selectively deposits binder onto the powder bed, corresponding to the cross-sectional slice of the CAD model. If color printing is enabled, CMYK inks are simultaneously jetted to produce full-color prototypes. The deposited binder causes the powder particles to adhere together in the defined regions, forming a solidified layer. After each pass, the build platform lowers by one layer thickness while the feed bed



risers to supply fresh powder, which is evenly spread across the surface by a roller. This sequence is repeated layer by layer until the entire part is completed within the surrounding powder bed.



2

### ***Hardware and Principle of Binder Jetting***

Once complete, the part remains surrounded by loose, unbound powder, which acts as a natural support. The surrounding powder is removed in a depowdering station using air brushes or vacuum. Post-processing such as infiltration, painting, or sanding may be applied depending on the intended application.

### **ADVANTAGES**

1. ***Low-cost and fast:*** Compared to other AM methods, parts can be printed quickly at relatively low cost.
2. ***Full-color capability:*** Unique ability to print parts in full color using CMYK binder inks.
3. ***No support structures:*** Loose powder provides built-in support, simplifying the process.
4. ***Room-temperature process:*** No lasers, heaters, or inert atmospheres required.
5. ***Educational and visualization use:*** Excellent for concept models, design reviews, and demonstration purposes.

### **LIMITATIONS**

1. ***Low mechanical strength:*** Parts are fragile and unsuitable for functional testing without infiltration.
2. ***Surface porosity:*** Parts are porous and require post-processing for durability or smooth finishes.
3. ***Limited material options:*** Primarily gypsum-based powders, with restricted mechanical performance.

4. **Post-processing dependency:** To achieve durability, infiltration with resins or wax is usually required.
5. **Not suitable for end-use parts:** The process is intended for concept modelling and visualization, not production.

### 2.1.6 3DP MATERIALS

#### 1. BUILD MATERIAL (In Powder Form):

- Plaster/Gypsum-based powders: The most common build material used by Z Corp printers. These powders are inexpensive, easy to spread, and suitable for absorbing binder.
- Starch-based powders: In earlier systems, starch powders were also used due to their compatibility with water-based binders.
- Composite powders: Some proprietary blends were developed to improve green strength and surface finish.

#### 2. BINDER:

- Typically, a water-based liquid binder jetted through an inkjet printhead.
- Contains additives such as polymers, sugars, or salts that promote adhesion between powder particles.

#### 3. COLOURING AGENTS:

- CMYK inks can be added to the binder, enabling full-color 3D printing.

#### 4. POST-PROCESSING / INFILTRANTS:

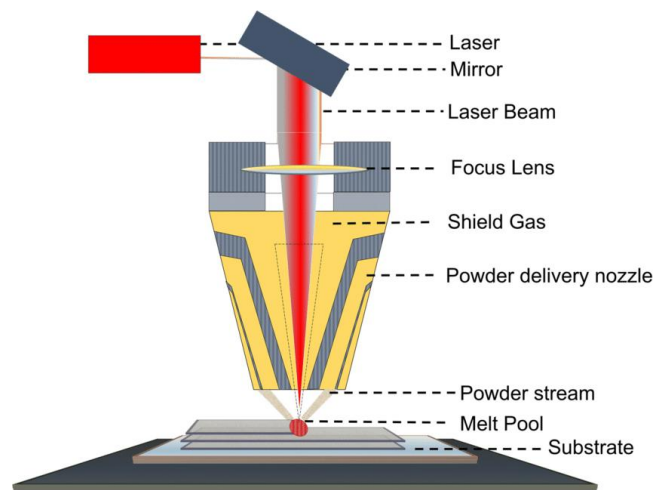
- Cyanoacrylate (super glue): Commonly used to infiltrate parts and give them strength.
- Epoxy resins: Provide higher durability and smoother finishes.
- Wax or paraffin: Used for sealing porous parts.

### 2.1.7 LASER ENGINEERED NET SHAPING (LENS)

Laser Engineered Net Shaping (LENS) is a process in which near-net-shape metal structures are built of powdered metal layer by layer from CAD design. LENS is a promising metal additive manufacturing process. It prints metal parts using a variety of materials: Cobalt, Aluminum, Stainless steel, Copper Titanium, Nickel and Tool steel using a high-powered Nd:YAG laser. The LENS process is originally developed at Sandia National Laboratories, LENS has been refined and commercialized by Optomec, a privately-held US based company headquartered in New Mexico.

#### PRINCIPLE

- In LENS, metal powder is fed through nozzles into a molten pool created by a high-power laser on the substrate or previously deposited layer.
- The laser melts the powder as it is deposited, allowing it to solidify immediately and form a dense, near-net-shape part.
- Argon gas is used to deliver the powder and protect the melt pool from contamination.



***Principle of LENS***

**HARDWARE:**



***Hardware of LENS***

1. Laser Unit and Laser Head: LENS machine uses Nd: YAG (Neodymium: Yttrium Aluminium Garnet) laser of 500W – 3kW. The laser head is surrounded by four equi-spaced nozzles through which the powder is deposited exactly at the point where the laser beam is focused. The laser head move in Z-axis up and down.
2. Build Chamber: This chamber is hermetically sealed and is connected to the glove-box with a small door. The system provides two rubber gloves for handling powder/ components and making required arrangements inside the chamber.
3. Build Platform: A platform, rectangular in shape, is provided in the build chamber for horizontal X-Y directional movement. The platform is slotted with threaded holes to attach the substrate of various sizes.

4. Build Substrate: The plate that is used in-between deposition head and build platform is called the substrate. A 2mm thick plate of appropriate material is adequate to be used as substrate.
5. Argon Gas Circulator: Most of the metals tend to oxidize at higher temperatures. To mitigate the issue of oxidation, argon gas needs to be circulated throughout the material deposition process.
6. Powder Hoppers: The LENS machine has two hoppers with a feeder mechanism to supply the powder to the nozzles from the hoppers. The glass doors are used to monitor the quantity of powder in the hoppers.

#### PROCESS:

- In LENS system the metal powders are delivered and distributed around the circumference of the head either by gravity or by using an inert, pressurized carrier gas, which also shields the melt pool from atmospheric oxygen.
- The metal powder in a slurry generated by the action of a high power beam laser is deposited on the upper surface of a metal substrate previously deposited on a platform.
- The build platform surface moves in XY direction to produce each layer of the object individually. The head moves vertically up each time a single layer is made.
- The part is built layer by layer under the control of software. When complete, the part is removed and can be heat-treated, Hot-Isostatic Pressed, machined, or finished.

#### ADVANTAGES

1. **High precision and complex geometries**: Can produce near-net-shape metal parts with intricate designs.
2. **Material flexibility**: Supports a wide range of metals and alloys.
3. **Repair and additive capability**: Can repair worn or damaged components by adding material only where needed.
4. **Minimal material wastage**: Material is deposited only where required, reducing scrap.
5. **Rapid prototyping and small-batch production**: Suitable for fast manufacturing of functional parts with good mechanical properties.

#### DISADVANTAGES

1. **High equipment cost**: LENS machines are expensive to purchase and maintain.
2. **Surface finish issues**: Parts may have rough surfaces and often require post-processing.
3. **Lower deposition rate for large parts**: Slower compared to wire-fed DED for bulk manufacturing.
4. **Skilled operation required**: Precise control of parameters is necessary to avoid defects.
5. **Risk of porosity and residual stresses**: Improper process control can lead to internal defects in parts.

#### **2.1.8 LENS MATERIALS**

1. Titanium and Titanium Alloys: Lightweight, high strength, corrosion-resistant; widely used in aerospace and biomedical implants.

2. Stainless Steel: Strong, corrosion-resistant; used for structural components, tooling, and industrial parts.
3. Nickel-based Superalloys: High-temperature strength and oxidation resistance; ideal for turbine blades, aerospace, and energy applications.
4. Cobalt-Chrome Alloys: High wear resistance and biocompatibility; used in medical implants and dental applications.
5. Aluminum Alloys: Lightweight with good strength-to-weight ratio; used in aerospace and automotive components.

#### 2.1.9 APPLICATIONS OF LENS

1. **Aerospace Components**: Manufacturing and repair of turbine blades, engine parts, and structural components.
2. **Biomedical Implants**: Customized bone implants, prosthetics, and dental components.
3. **Tooling and Molds**: Production of complex molds, dies, and repair of worn-out tools.
4. **Repair and Maintenance**: Restoration of high-value or worn-out metal parts in industries like aerospace, automotive, and energy.
5. **Rapid Prototyping**: Creation of functional prototypes and small-batch production of metal parts.

***LAMINATED OBJECT MANUFACTURING***

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**COURSE CONTENT**

- 2.2.1 Introduction**
- 2.2.2 Laminated Object Manufacturing (LOM)**
- 2.2.3 LOM Materials**
- 2.2.4 Applications**

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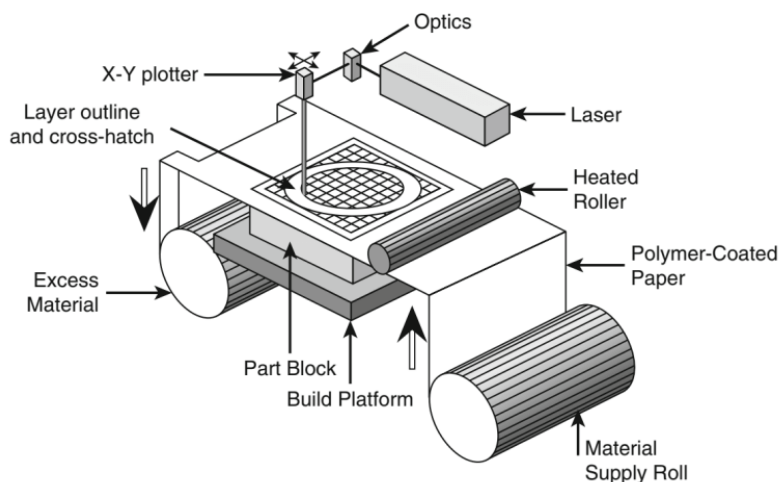
### 2.2.1 INTRODUCTION

In Section 1.3: Solid-Based Processes, Lamination-Based Processes were introduced. In this section, the Laminated Object Manufacturing (LOM) process is discussed in detail.

Several other processes have been developed based on the principle of sheet lamination, utilizing different build materials and cutting strategies. Due to the nature of their construction, only the outer contours of the parts are typically cut. The process sequence may vary sheets can either be cut first and then stacked (or) stacked first and then cut.

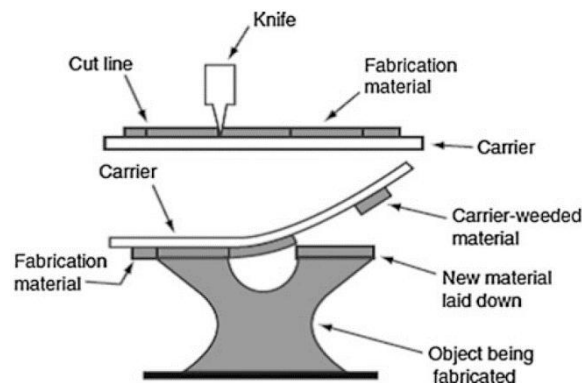
Among the various sheet lamination techniques, one of the most popular methods employs paper as the build material, bonded using a polymer-based adhesive. The typical paper thickness used ranges from 0.07 mm to 0.2 mm. Sheet Lamination process is also referred as Layer Laminate Manufacturing. The sheet lamination processes are classified as

1. **Bond-Then-Form Processes:** Sheet material layers are bonded together before cutting them into the desired shape to create a 3D geometry.



***Bond-Then-Form Processes***

2. **Form-Then-Bond Processes:** sheet material is cut to shape first and then bonded to the substrate. This approach is popular for construction of parts in metallic or ceramic materials that are thermally bonded.



***Form-Then-Bond Processes***

### 2.2.2 LAMINATED OBJECT MANUFACTURING (LOM)

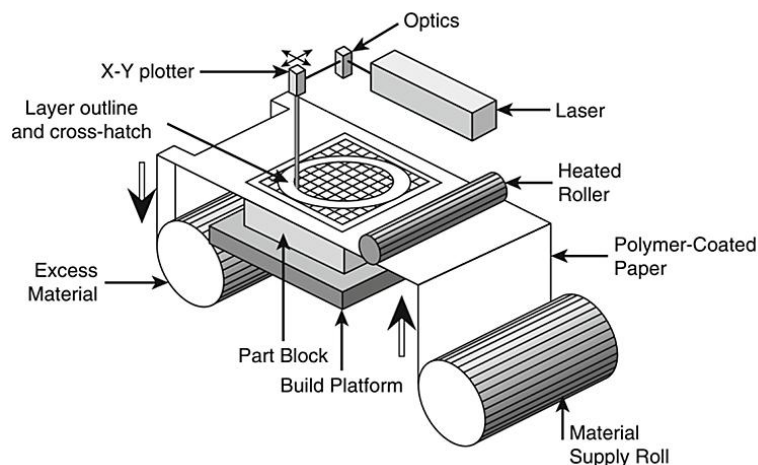
LOM is a solid-based additive manufacturing process originally developed by Helisys Inc. (now Cubic Technologies). The LOM process fabricates 3D models by laminating layers



of adhesive-coated sheet material typically paper (or) plastic foils and cutting each layer to shape using a laser. The process produces parts with good surface finish and rigidity, suitable for concept modeling, tooling, and pattern making.

### HARDWARE

- Build Platform: The platform serves as the base on which the part is fabricated layer by layer. After each layer is completed, the platform moves downward in the Z-direction to allow deposition and lamination of the next sheet.
- Material Supply and Take-up Rolls: The build material is provided in the form of continuous sheets wound on a supply roll. Fresh material is drawn from this roll, laminated onto the build platform, and after cutting, the take-up roll collects the waste or excess sheet material.
- Laminating Roller: A heated roller applies pressure and heat to bond the newly fed sheet to the previously laminated layer. This ensures strong adhesion and minimizes delamination during cutting.
- Laser Cutting System: A CO<sub>2</sub> laser is used to cut the cross-sectional outline of the layer according to CAD data, as well as to create crosshatch patterns in surrounding waste areas for easy removal after fabrication.



***Hardware and Principle of LOM***

- The process begins with the CAD model of the part, which is converted into an STL file. The software slices the model into thin layers and generates tool paths for cutting each contour.
- A fresh sheet of adhesive-coated material is fed onto the build platform. The heated roller bonds it to the previous layer. The laser then cuts the outline of the part's cross-section and hatches the excess material into a grid pattern.
- After each layer is cut, the platform lowers by one sheet thickness, and the next sheet is advanced, laminated, and cut. This process repeats until all layers are stacked and bonded.
- The final output is a solid block containing the fully enclosed part. The surrounding crosshatched waste material is manually removed to extract the model. The resulting wood-like LOM part can be finished using traditional post-processing methods such as sanding, polishing, painting, or sealing.

ADVANTAGES:

1. **Wide Range of Materials:** LOM can utilize a diverse range of organic and inorganic sheet materials, including paper and plastic films, making it a versatile process for various prototyping needs.
2. **High Build Speed:** Since the laser only traces the outer contour of each layer rather than scanning the entire surface, LOM offers faster build times, especially for parts with large, solid cross-sections.
3. **Good Dimensional Accuracy:** LOM systems typically achieve a dimensional accuracy better than  $\pm 0.127$  mm, providing sufficient precision for conceptual prototypes.
4. **No Need for Separate Support Structures:** The surrounding uncut material naturally supports the part during fabrication, eliminating the need for additional support structures.
5. **Cost-Effective for Large Models:** The use of inexpensive materials such as paper and relatively fast build rates makes LOM a cost-effective solution for producing large, non-functional prototypes and casting patterns.

LIMITATIONS:

1. **Unsuitable for Thin-Walled Features:** The process is not ideal for producing delicate or thin-walled geometries, particularly along the Z-axis.
2. **Limited Mechanical Strength:** Since layers are bonded primarily through adhesive and thermal lamination, the interlayer strength is relatively low. As a result, LOM parts are generally not suitable for load-bearing applications.
3. **Labor-Intensive Post-Processing:** After fabrication, the part must be manually separated from the surrounding crosshatched waste material, often using tools similar to wood-carving instruments.
4. **Surface Finish and Anisotropy:** The final part often exhibits a layered wood-grain appearance and may require additional finishing operations such as sanding or sealing.

### 2.2.3 LOM MATERIALS

LOM utilizes sheet-based feedstock materials that are adhesive-coated and bonded layer-by-layer through heat and pressure. The choice of material depends on the intended application, desired mechanical strength, surface finish, and post-processing requirements.

Commonly used Materials are:

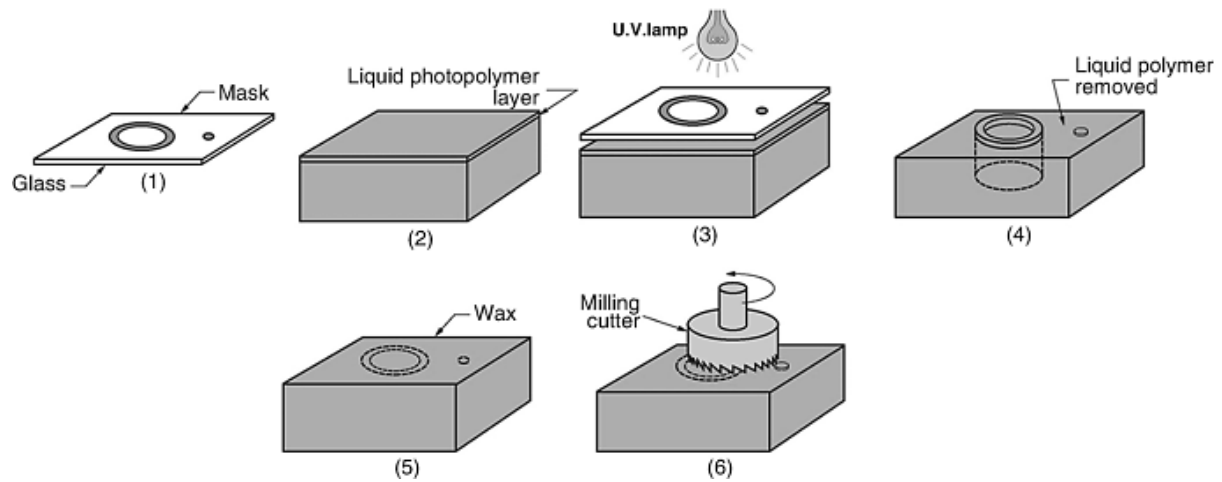
1. Paper Sheets: The most used material in LOM is paper sheets, which produce parts with a wood-like appearance and texture. These sheets typically have a thickness ranging from 0.07 mm to 0.2 mm and are bonded using a polymer-based adhesive. Paper-based LOM parts are particularly suitable for concept models and visual prototypes, offering good surface finish and dimensional stability at low cost.
2. Plastic Films: In addition to paper, plastic films such as Polyvinyl Chloride (PVC), Polycarbonate (PC), and Polypropylene (PP) are also used in LOM systems. These materials provide greater durability, improved moisture resistance, and smoother surfaces compared to paper-based builds, making them suitable for applications that require enhanced strength and surface quality.

#### **2.2.4 APPLICATIONS**

LOM is primarily used for conceptual modeling, aesthetic visualization, and tooling applications, where mechanical performance is less critical than geometry and appearance.

- Conceptual and Design Prototypes: Used for visual models, form and fit testing, and design verification in early stages of product development. Offers good surface finish and dimensional accuracy for large-scale parts.
- Casting and Tooling Patterns: Ideal for producing patterns for sand casting, investment casting, and vacuum forming molds. Paper or composite LOM parts can be infiltrated or coated to improve strength and surface hardness.
- Architectural and Display Models: Used to create architectural mock-ups, scale models, and exhibition prototypes due to low cost and fast fabrication of large geometries.
- Educational and Research Models: Useful in academic and training environments to demonstrate additive manufacturing principles and to create geometric prototypes.

## ***SOLID GROUND CURING***



### **COURSE CONTENT**

- 2.3.1 Introduction**
- 2.3.2 Machine Details**
- 2.3.3 Principle**
- 2.3.4 Applications**

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### 2.3.1 INTRODUCTION

Solid Ground Curing (SGC) is a resin-based rapid prototyping process manufactured by Cubital in Israel. SGC materials are generally UV-curable photopolymers, similar to those used in stereolithography, but formulated for high-speed, full-layer exposure. These materials are typically acrylate or epoxy-based resins with additives that enhance curing speed, dimensional stability, and mechanical strength. Some formulations may include fillers or pigments to improve surface finish and reduce shrinkage. The solidified resin layers are supported by wax material, which fills the non-cured regions during the process and is later melted away, resulting in parts with excellent dimensional accuracy, smooth surface finish, and minimal post-processing requirements.

### 2.3.2 MACHINE DETAILS

The SGC systems are physically quite large, set up in an automatic assembly line fashion with the curing lamp, resin vacuum, milling station, etc. They are currently offered in two systems, the entry level SGC 4600 and the high-end SGC 5600. The SGC 4600 systems offers a 350mm x 350 mm x 350mm (14" x 14" x 14") working envelope, whereas the SGC 5600 system has a 500mm x 350mm x 500mm (20" x 14" x 20") envelope and a 300% increase in net production rate over the SGC 4600 system.

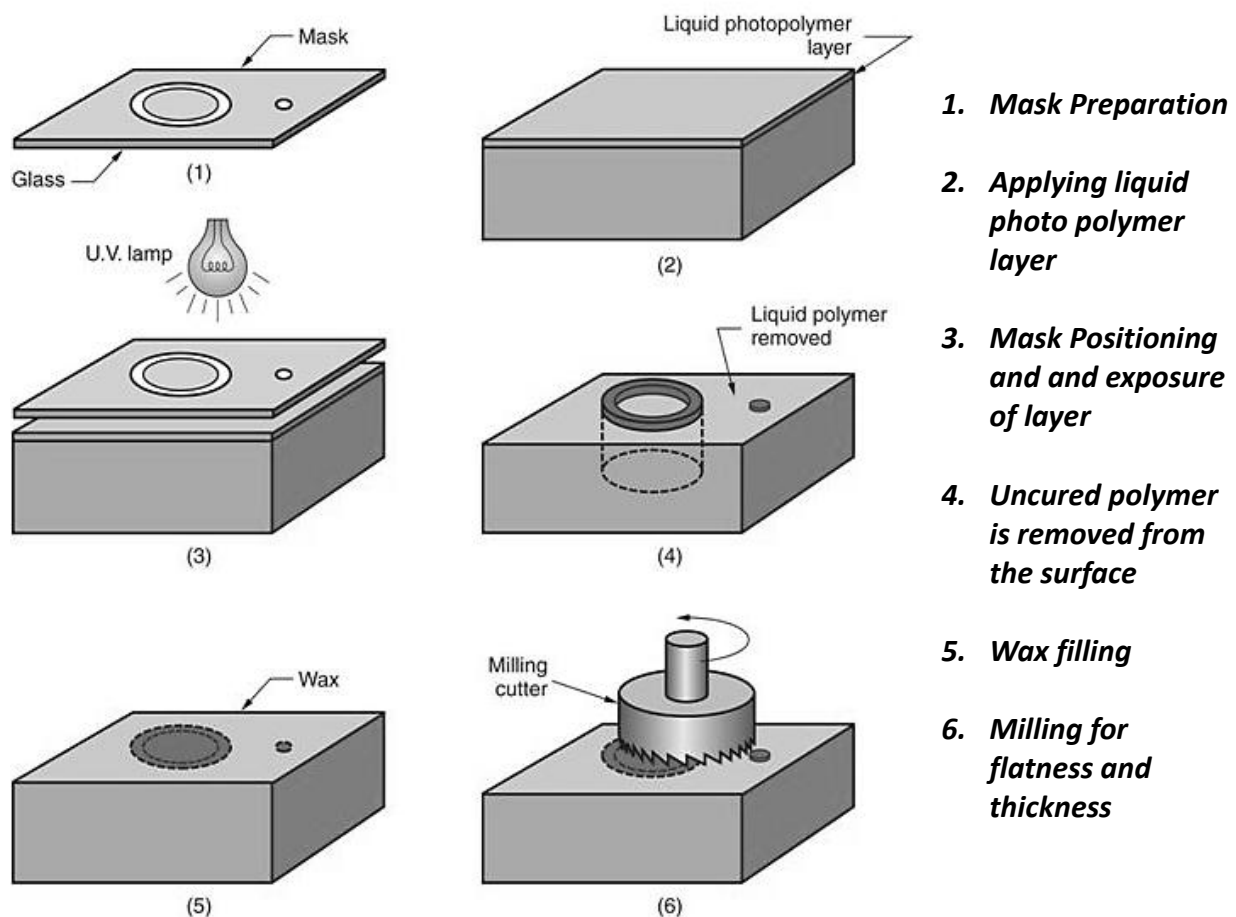
*Specifications of SGC Machines*

| Model                                 | Solider 4600                                | Solider 5600                                |
|---------------------------------------|---------------------------------------------|---------------------------------------------|
| Irradiation medium                    | High power UV lamp                          |                                             |
| XY resolution (mm)                    | Better than 0.1                             |                                             |
| Surface definition (mm)               | 0.15                                        | 0.15                                        |
| Elevator vertical resolution (mm)     | 0.15                                        | 0.1–0.2                                     |
| Minimum feature size (mm)             | 0.4 (horizontal, X–Y)<br>0.15 (vertical, Z) | 0.4 (horizontal, X–Y)<br>0.15 (vertical, Z) |
| Work volume, XYZ (mm × mm × mm)       | 350 × 350 × 350                             | 500 × 350 × 500                             |
| Production rate (cm <sup>3</sup> /hr) | 550                                         | 1311                                        |
| Minimum layer thickness (mm)          | 0.06                                        | 0.06                                        |
| Dimensional accuracy                  | 0.1%                                        | 0.1%                                        |
| Size of unit, XYZ (m × m × m)         | 1.8 × 4.2 × 2.9                             | 1.8 × 4.2 × 2.9                             |
| Data control unit                     | Data Front End (DFE) workstation            |                                             |
| Power supply                          | 380–415 V <sub>AC</sub> , 3 phase, 50 kW    | 380–415 V <sub>AC</sub> , 3 phase, 50 kW    |

### 2.3.3 PRINCIPLE

SGC process is based on the principle of layer-by-layer photopolymer solidification using a high-intensity ultraviolet (UV) light source combined with a mask projection system for full-layer exposure. The main principles of the process are as follows:

1. Parts are built layer by layer from a liquid photopolymer resin that solidifies when exposed to UV light. Unlike stereolithography, which uses a laser to trace each cross-section, SGC employs masked illumination, where the entire layer is exposed simultaneously. The image of each layer is generated by a non-impact ionographic printing process, similar to the xerographic process used in photocopiers.
2. Multiple parts can be processed simultaneously by grouping them into batches. This capability enhances productivity by enabling parallel fabrication of several components within the same build cycle.
3. Each layer of the build contains cross-sectional slices of one or more parts. All slices in a single layer are created simultaneously, ensuring high throughput. Layers are intentionally formed slightly thicker than required, and after curing, a milling operation is performed to achieve precise layer thickness and excellent vertical accuracy.
4. The SGC process is self-supporting, eliminating the need for additional support structures. The unexposed regions of each layer are filled with meltable wax, which solidifies and provides mechanical support during fabrication. Once the build is complete, the wax is melted or dissolved away, revealing the finished part with smooth surfaces and accurate dimensions.



### *Principle of SGC*

#### PROCESS

The SGC process begins with the preparation of a 3D CAD model of the part to be produced. The model is digitally sliced into thin cross-sectional layers using Cubital's Solider

DFE (Data Front End) software, which verifies, corrects, and converts the CAD file into an acceptable format such as STL. The processed slice data are then transferred to the mask generation unit, where an electrostatic charge image of each layer is created on a transparent mask plate through a non-impact ionographic process, similar to xerographic printing. The charged image is developed using black electrostatic toner to form an opaque pattern that represents the solid regions of the part.

Once the mask is generated, a thin layer of liquid photopolymer resin is spread uniformly over the build surface. The mask is positioned above the resin layer, and the entire surface is exposed to a collimated UV light source, curing the exposed regions corresponding to the layer geometry. The unexposed resin remains in liquid form and is removed by vacuum suction. The resulting voids are then filled with molten wax, which solidifies upon cooling to provide complete mechanical support. The layer is then milled to an exact thickness, producing a flat, smooth surface ready for the next layer.

This sequence of operations resin coating, masked UV exposure, resin removal, wax filling, cooling, and milling is repeated automatically until the final layer is completed. In advanced SGC systems such as the Cubital Solider 5600, the part undergoes an additional UV flood-curing step after each layer to ensure complete polymerization.

Once all layers are finished, the supporting wax is melted or dissolved away, leaving behind a fully solid, dimensionally accurate, and smooth-surfaced prototype.

#### ADVANTAGES

1. **Parallel processing:** The process is based on instant, simultaneous curing of a whole cross-sectional layer area (rather than point-by-point curing). It has a high speed throughput. Many parts can be produced at the same time.
2. **Self-supporting:** The solid wax supports the part in all dimensions and therefore a support structure is not required.
3. **Job changing is easier:** Removable trays allow job changing during a run and layers are erasable.
4. **Unique part properties:** The part that the Solider system produces is reliable, accurate, sturdy, machinable, and can be mechanically finished.
5. **Minimum shrinkage effect:** This is due to the full curing of every layer.
6. **High structural strength and stability:** This is due to the curing process that minimizes the development of internal stresses in the structure. As a result, they are much less brittle.
7. **No hazardous odors are generated:** The resin stays in a liquid state for a very short time, and the uncured liquid is wiped off immediately. Thus, safety is considerably higher.

#### DISADVANTAGES

1. **Requires large physical space:** The size of the system is much larger than other systems with a similar build volume size.
2. **Wax gets stuck in corners and crevices:** It is difficult to remove wax from parts with intricate geometry. Thus, some wax may be left behind.
3. **Waste material produced:** The milling process creates shavings, which have to be cleaned from the machine.

4. **Noisy:** The Solider system generates a high level of noise as compared to other systems.

### 2.3.4 APPLICATIONS

The SGC process became obsolete due to its high system complexity, expensive equipment, and slow overall build rate. Although it could cure entire layers simultaneously, the additional steps such as waxing, cooling, and milling made it time-consuming and costly to operate. The process also required frequent maintenance, occupied large space, and consumed significant energy. With the emergence of simpler, faster, and more economical technologies like SLA, SLS, FDM, and PolyJet printing, SGC gradually lost its relevance and is discontinued.

However, following are some of the applications

1. **Functional Prototypes:** Used to create durable and highly accurate prototypes for functional testing in industries such as automotive, aerospace, and consumer products.
2. **Tooling and Molds:** Ideal for producing precise molds, patterns, and tooling inserts due to its excellent dimensional accuracy and superior surface finish.
3. **Casting Patterns:** Commonly applied in manufacturing investment casting and sand casting patterns where fine details and smooth surfaces are essential.
4. **Complex Geometries:** Suitable for fabricating parts with intricate internal structures, undercuts, and overhangs, effectively supported by wax during the build process.